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Transferable Model-Based Reinforcement Learning for Vehicular Platoon Control

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***Abstract*—The learning efficiency remains a critical impediment to the practical application of connected and automated vehicles (CAVs). This paper proposes a transferable model-based reinforcement learning (TMBRL) strategy to enhance the sample efficiency and learning rate of CAVs. Specifically, a surrogate policy model is established by capturing state transitions between the actual environment and the vehicle within traffic scenarios. Then, a model-based reinforcement learning (MBRL) approach is established utilizing a surrogate model and a soft actor-critic algorithm. To improve the learning efficiency of platoon control algorithm, a transfer learning method is implemented to MBRL framework. Specifically, the trained surrogate model of vehicles in the source domain is transferred to vehicles in the target domain, and the latter just should update the surrogate model in terms of the individual dynamic characteristics and tasks. Finally, a platoon experiment platform with Prescan software is conducted. The experimental evaluation demonstrates that the TMBRL strategy significantly outperforms conventional reinforcement learning approaches, achieving higher average cumulative reward of 47 and demonstrating a 16% improvement in training success rate. Comparative analysis further reveals that the proposed TMBRL strategy exhibits superior robustness in platoon tracking tasks, maintaining enhanced trajectory tracking precision and stability under dynamic environmental conditions.**

***Index Terms*—Vehicular platoon, reinforcement learning, transfer learning, surrogate model.**

I. Introduction

THE evolution of connected and autonomous vehicles (CAVs) has experienced a profound surge, driven by advancements in artificial intelligence and wireless communication [1], [2]. The vehicular platoon involves the synchronization of CAVs to travel in complex traffic scenarios [3]. Achieving satisfied learning efficiency is critical to enabling the seamless processing of massive data streams while maintaining the robust safety of CAVs [4].

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Among numerous learning-based approaches, reinforcement learning (RL) stands out for its significant achievements in CAVs [5], [6]. Compared to conventional control approaches, reinforcement learning method enables vehicles to autonomously derive optimal coordination strategies through continuous interaction with their environment. The RL agent could be trained using data on vehicle states and environmental factors (e.g., traffic and road conditions) [7]. Actually, RL is categorized into model-based reinforcement learning (MBRL) and model-free reinforcement learning (MFRL). Model-based reinforcement learning uses an explicit environment model for planning, enabling sample-efficient learning. In contrast, model-free methods learn policies directly from experience without modeling the environment, often requiring more data and exhibiting slower convergence. Several applications utilizing MBRL have emerged within the domain of vehicle control [8], [9], [10]. It should be pointed out that the offline training process suffers from higher learning complexity to construct and refine the model [11]. A reason behind this phenomenon is that the interactions between the agent and environment are inefficient, resulting in each iteration yielding only a minimal amount of data. To mitigate the abovementioned problem, the concept of MBRL has received extensive attention [12]. MBRL seeks to develop a virtual surrogate model with acceptable accuracy by capturing state transitions between the actual environment and the agent. Specifically, this model is capable of predicting state transitions according to the actions taken by the RL agent, producing supplementary interaction data and speeding up the training process. MBRL can achieve superior sample efficiency compared to MFRL as it relies on the Markov decision process (MDP) model representing the state transition dynamics of the environment. Therefore, the MBRL method has been extensively utilized in industrial automation [13], [14]. Recently, a model-based deep reinforcement learning method without sacrificing safety is presented to ensure the stability of single vehicle [15]. Hu et al. developed a safe model-based RL algorithm to solve the collision-free model-reference trajectory tracking problem of CAVs [16]. Nevertheless, the model-based reinforcement learning approach has yet to be applied to vehicular platoon control, and several inherent challenges, including the accuracy of the system model and real-time learning efficiency, require further exploration.

Actually, the learning efficiency is hindered by the limited computational power of individual vehicles, which struggle to meet the real-time processing needs of large-scale platooning systems [17]. The computational complexity associated with

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managing vehicle platoons poses a significant challenge in the integration of advanced control and coordination algorithms, especially given the constraints of resource-limited onboard vehicular systems [18]. As the size of the vehicle platoon expands, the computational cost of MBRL also increases proportionally, posing challenges for implementing the RL scheme on vehicular chips. Therefore, several computational optimization methods such as parallel computing [19], batch processing [20], edge computing [21] and numerical approximation [22] were proposed. However, these methods often overlook the heterogeneous characteristics of vehicular platoon. In particular, the diverse dynamics and capabilities of various vehicle types along with their distinct response times, processing abilities and resource constraints. Here, the transfer learning method provides a feasible solution for mitigating computational burdens of RL scheme [23], [24]. The key principle behind transfer learning in RL is to provide available knowledge for the target task, thus accelerating the learning process and reducing computational costs [25]. This technique leverages knowledge from the source domain to improve convergence performance in similar target domains. Recent research has explored selective layer freezing and task-specific subnetwork extraction to minimize unnecessary computations while maintaining transfer performance [26]. The majority of the literature on transfer RL primarily addresses policy and reward transfer, rather than focusing on model transfer [27]. A key challenge in advancing this field toward a model-driven approach is achieving domain adaptation for the dynamic modules. Therefore, how to develop an efficient control strategy with transfer learning for vehicular platoons is an ongoing research topic.

The research mentioned above in platoon control methods focuses on MFRL, which relies on extensive environment interactions for learning, leading to slower convergence and lower sample efficiency. Furthermore, the existing transfer learning method focuses on policy and reward transfer, while research on model transfer remains insufficient. All these factors motivate us to construct an efficient MBRL scheme for vehicular platoon. The main contributions of this paper are summarized below.

i) An MBRL algorithm is developed to alleviate the dependency on the complex traffic environment. While model accuracy is an inherent challenge in reinforcement learning, the primary contribution of our approach lies in applying MBRL to vehicular platoon control. It utilizes a learned model to predict future states and rewards for different actions, thereby improving policy optimization efficiency. Different from traditional RL methods that rely on static models or fixed policies [7], [9], our approach enhances policy optimization efficiency by continuously updating the model to reflect changes in the traffic environment for vehicular platoon.

ii) A transfer learning approach is designed to improve the computational efficiency of policy learning. Compared to traditional lightweight methods require extensive retraining across all vehicles [18], [21], our proposed approach transfers knowledge from vehicles in a source domain to those in a target domain. The target vehicles only need to update their individual dynamic characteristics and task-specific parameters, significantly reducing the computational burden compared to existing methods. This approach provides a more scalable solution for policy learning in dynamic environments, outperforming conventional techniques by reducing both training time and resource consumption.

The remainder of this paper is organized as follows. Section II introduces the necessary preliminaries, and Section III presents the design of the proposed strategy. Section IV then describes the field experiments, while Section V analyzes the experimental results. Finally, Section VI concludes the study.

II. Preliminaries

A. Vehicular Platoon Model

The vehicular platoon consists of M CAVs with the predecessor leader following communication topology. To formulate a simplified model, the following assumptions are adopted: 1) The vehicle operates on a dry, level road surface, with negligible longitudinal tire slip. 2) The vehicle's body is assumed to be symmetric and rigid [28]. Then, the dynamics of vehicle $i(i = 1, 2, \cdots M)$ are described as

$$\begin{cases} \dot{s}_i(t) = v_i(t) \\ \dot{v}_i(t) = \dfrac{1}{m_i}\left(\eta_i \dfrac{T_i(t)}{R_i} - C_{A,i}v_i^2 - m_i g_i f_i\right) \\ \tau_i \dot{T}_i(t) + T_i(t) = T_{i,des}(t) \end{cases} \tag{1}$$

where $s_i(t)$ and $v_i(t)$ represent the position and velocity of vehicle i, respectively. m_i denotes the vehicle mass. $C_{A,i}$ stands for the lumped aerodynamic drag coefficient. g_i is the acceleration due to gravity. f_i denotes the coefficient of rolling resistance. $T_i(t)$ and $T_{i,des}(t)$ represent actual and desired driving torque, respectively. τ_i stands for the inertial delay. R_i denotes the tire radius.

The feedback linearization technique is employed to transform the nonlinear model into a linear form. That is

$$\tau_i \dot{a}_i(t) + a_i(t) = u_i(t) \tag{2}$$

where a_i denotes the acceleration. u_i is the control input after linearization, which denotes the desired acceleration.

Each vehicle in the platoon should maintain a desired inter-vehicular spacing h^* and velocity v^*, where $h_i^*(t) = h_i(t) - \triangle h_i(t)$, $h_i(t) = s_{i-1}(t) - s_i(t)$ and $v_i^*(t) = v_i(t) - \triangle v_i(t)$. Then, the platoon error model with a desired equilibrium state (h^*,v^*) is defined as

$$\begin{cases} \triangle \dot{h}_i(t) = \triangle v_{i-1}(t) - \triangle v_i(t) \\ \triangle \dot{v}_i(t) = a_{i-1}(t) - a_i(t) \end{cases} \tag{3}$$

According to (1)-(3), the platoon error system is derived as the discrete form

$$\begin{aligned} &x_i(k+1) = A_i x_i(k) + B_i u_i(k) + D_i a_{i-1}(k) \\ &y_i(k+1) = C_i x_i(k) \\ &A_i = \begin{bmatrix} 1 & -T_s & -\tau_i \\ 0 & 1 & T_s \\ 0 & 0 & (\tau_i - T_s)\tau_i \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 0 \\ T_s/\tau_i \end{bmatrix}, \\ &D_i = \begin{bmatrix} 0 \\ -T_s \\ 0 \end{bmatrix}, \end{aligned} \tag{4}$$

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where $x_i(k) = [\triangle h_i(t), \triangle v_i(k), a_i(k)]^T$. C_i is an identity matrix and y denotes the output of the system states. T_s stands for the sampling period. a_{i-1} is regarded as a disturbance.

B. Reinforcement Learning With MBPO

Model-based policy optimization in reinforcement learning plays a crucial role in vehicle platoon control by enabling efficient decision-making through predictive models. It enhances scalability and adaptability, reducing computation costs while improving real-time control in dynamic traffic environments [10]. Model-based policy optimization (MBPO) integrates model-based and model-free reinforcement learning principles to achieve a balance between sample efficiency and policy performance. The core idea of MBPO lies in leveraging a learned dynamics model to generate short-horizon simulated rollouts, which augment the real environment data used for policy training. Soft Actor-Critic (SAC) serves as the core of MBPO, enabling the agent to learn a stochastic policy.

The interaction between the RL agent and the environment is modeled as a MDP, denoted by $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$. Specifically, $\mathcal{S}$ denotes the state space that comprises the state variable s. $\mathcal{A}$ is the action space, which includes candidate actions a. The state transition function is denoted by $\mathcal{P}$. Note that $r : \mathcal{S} \times \mathcal{A}$ indicates a reward function, and the discount factor is represented by γ.

To improve the sample efficiency in complex state and action spaces, MBPO strives to maximize the accumulated reward. The objective function of SAC is defined as

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(s_t,a_t)\sim\rho_\pi} \left[r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot|s_t)) \right] \tag{5}$$

where s_t and a_t represent the state and the action, respectively. For a stochastic policy $\pi : \mathcal{S} \times \mathcal{A} \to [0, +\infty)$, $\pi(\cdot|s_t)$ denotes the distribution of actions at state s_t. $\rho_\pi(s_t, a_t)$ is the state-action marginal of the trajectory distribution ρ_π. $\mathcal{H}(\pi(\cdot|s_t)) = -\mathbb{E}_{a_t\sim\pi}[\ln(\pi(a_t|s_t))]$ stands for the entropy of π, and $\alpha > 0$ represent the temperature parameter.

Let Q_λ and π_θ denote the critic network and the actor network, where λ and θ are the respective trainable parameters. Then, the parameter λ is updated by minimizing a soft Bellman residual, which is defined as

$$J_Q(\lambda) = \mathbb{E}_{(s_t,a_t)\sim\mathcal{B}} \left[\frac{1}{2} \left(Q_\lambda(s_t, a_t) - r(s_t, a_t) - V_\lambda(s_{t+1}) \right)^2 \right] \tag{6}$$

where $\mathcal{B} = \{s_t, a_t, r_t, s_{t+1}\}$ is the replay buffer. $V_\lambda(s_{t+1}) = \gamma Q_\lambda(s_{t+1}, a_{t+1}) - \gamma\alpha \ln(\pi_\theta(a_{t+1}|s_{t+1}))$.

Let Q_λ denote the target critic network, parameterized by $\bar{\lambda}$, while a_{t+1} is drawn from the action distribution $\pi_\theta(\cdot \mid s_{t+1})$. For the actor network π_θ, the parameter θ is optimized by minimizing the following objective function

$$J_\pi(\theta) = \mathbb{E}_{s_t\sim\mathcal{B}, a_t\sim\rho_\pi} \left[\alpha \log(\pi_\theta(a_t|s_t)) - Q_\lambda(s_t, a_t) \right]. \tag{7}$$

Furthermore, the temperature parameter α should be updated, seeking to align the actual entropy of the policy with the predefined target entropy. By minimizing this objective the training mechanism enables the dynamic adjustment of α. And it facilitates an appropriate balance between exploration and exploitation in the action selection process.

C. Transfer Learning

Transfer learning significantly enhances vehicle platoon control by enabling rapid adaptation to diverse traffic conditions, improving model efficiency and robustness while reducing the need for extensive retraining on new scenarios [23]. For the distributed platoon control framework, each CAV should train an individual RL policy in terms of real-time traffic information. This task presents a significant computational challenge, particularly for large-scale vehicular platoon. Here, the transfer learning emerges as a pivotal strategy for enhancing learning efficiency by sharing features and transferring model parameters. This technique focuses on leveraging the knowledge obtained from the source domain ($\mathcal{D}_S$) to enhance task execution in the target domain ($\mathcal{D}_T$), thereby improving task performance in the target domain. The source domain refers to the initial set of vehicle data, typically collected under well-defined conditions or configurations, used for training the model. The target domain refers to a different vehicle, where the learned knowledge is applied. The primary goal of transfer learning in this setting is to adapt the model trained on the source domain to the target domain, improving its ability to handle new, unseen conditions with minimal retraining.

The domain $\mathcal{D}_S$ of platoon involving a feature space $\mathcal{S}_s$ and a marginal probability distribution function $\mathcal{P}(s)$ is described as

$$\mathcal{D}_S = \{\mathcal{S}_s, \mathcal{P}(s)\}, s \in \mathcal{S}_s. \tag{8}$$

Accordingly, the domain $\mathcal{D}_T$ comprises a label space $\mathcal{S}_r$ and a predictive function $\mathcal{F}(s)$, such that

$$\mathcal{D}_T = \{\mathcal{S}_r, \mathcal{F}(s)\}, s \in \mathcal{S}_r. \tag{9}$$

Given the differences in feature distributions between $\mathcal{D}_S$ and $\mathcal{D}_T$, it becomes essential to conduct classification and prediction on the unlabeled datasets within the target domain $\mathcal{D}_S$. Therefore, a Kullback-Leibler divergence method is utilized to measure the similarity between $\mathcal{D}_S$ and $\mathcal{D}_T$ [29]. This can be described by the following formula

$$\mathcal{D}_{KL}(P \parallel Q) = \sum_{s\in\mathcal{S}_r} \mathcal{P}(s) \ \log \frac{\mathcal{P}(s)}{\mathcal{F}(s)}. \tag{10}$$

III. TMBRL Strategy Design

Fig. 2 presents the framework of the proposed transferable model-based reinforcement learning (TMBRL) strategy for vehicular platoon. Specifically, the real-time platoon experiences are stored in replay memory and utilized to construct an environment model of the source domain vehicle CAV_1. This model generates additional experiential data, which is stored separately to enhance the learning process. SAC algorithm is employed as a foundational component in the construction of MBPO policy agent in reinforcement learning. Then, the trained knowledge involving the policy network and environment model is uploaded to the buffer, and then it is transferred to the target domain vehicle CAV_i $(i > 1)$.

A. Surrogate Model

The MBPO aims to construct a surrogate model $p_\psi(s_{t+1}|s_t, a_t)$ to simulate the interaction of environment,

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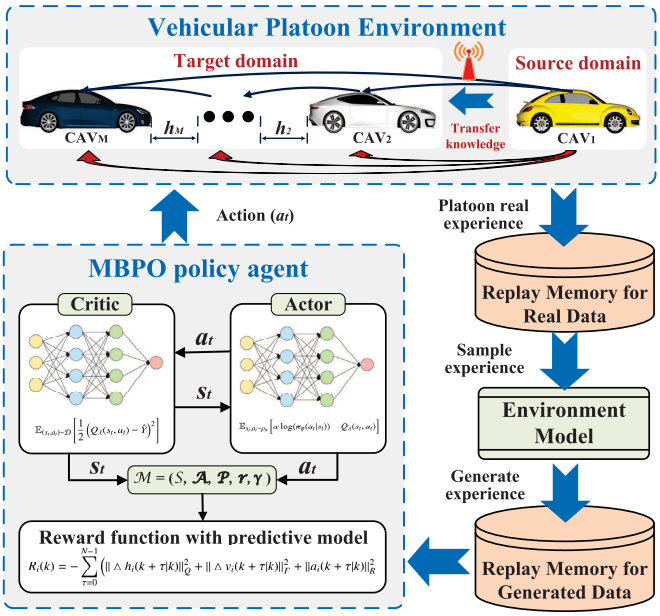

Fig. 1. Framework diagram of the TMBRL method.

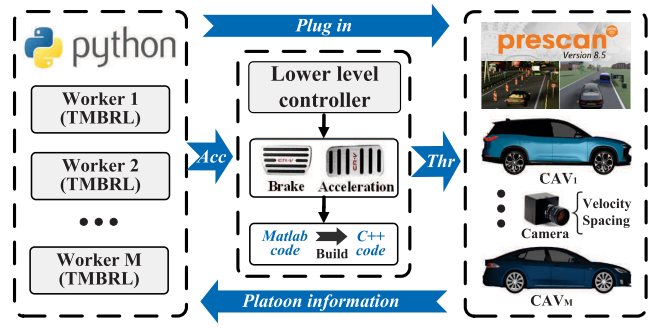

Fig. 2. Experimental framework.

allowing for the generation of synthetic data. Here, we learn a mapping model to directly represent the environment's transition and reward functions within a platoon.

With this in mind, the kinetic transition process of the environment could be trained as a network $p_\psi(s_t, a_t) = s_{t+1} - s_t$. To optimize the effectiveness of model training, it is advisable to minimize the prediction error

$$M_{single}(\psi) = \frac{1}{M} \sum \left\| (s_{t+1} - s_t) - p_\psi(s_t, a_t) \right\|_2 \tag{11}$$

where M denotes the size of the dataset.

Although (11) ensures the accuracy of model at a single step, preventing error accumulation over extended horizon rollouts remains a challenge. To mitigate this issue, a multistep loss function is designed to assess accumulated errors over time. The multistep loss function is described as

$$M_{multi}(\psi) = \frac{1}{M} \sum \sum_{\tau=1}^{L} \left\| (s_{t+\tau} - s_{t+\tau-1}) - p_\psi(s_{t+\tau-1}, a_{t+\tau-1}) \right\|_2^2 \tag{12}$$

where L stands for the prediction horizon.

Selecting the appropriate surrogate model is essential for effectively capturing the dynamics and uncertainties of the environment. Note that the surrogate model is attractive due to their adeptness in harmonizing advanced function approximation capabilities with pronounced sample efficiency, thus enhancing their applicability in complex traffic system. However, capturing the uncertainty within the model's prediction process remains a significant challenge. Therefore, an ensemble dynamics model is adopted to characterize environmental dynamics, where each member of the dynamics model is formed as a probabilistic network. Such that

$$p_\psi(s_{t+1}|s_t, a_t) = \mathcal{N}\left(\mu_\psi(s_t, a_t), \sigma_\psi^2(s_t, a_t)\right) \tag{13}$$

where $\mathcal{N}\left(\mu_\psi(s_t, a_t), \sigma_\psi^2(s_t, a_t)\right)$ denotes a Gaussian distribution parameterized by the mean of the distribution $\mu_\psi(s_t, a_t)$ and the variance of the distribution $\sigma_\psi^2(s_t, a_t)$.

The independent probabilistic model captures the uncertainty of the environment to generate its dynamic characteristics. By selecting a model at random from the probabilistic ensemble, we can produce environmental predictions that enable diverse transitions to be sampled across various dynamic models.

B. Reward Function

The design of the reward function is vital in MBRL for vehicle platooning applications. It could evaluate the actions of the agent, thus guiding the policy to achieve a balance among performance objectives.

In this part, the reward function involves driving safety objective and driving comfort objective. To be more specially, driving safety, denoted as $J_i^s = \| \triangle h_i(t+\tau|t)\|_Q^2$, pertains to the capability of mitigating accident risks and enhancing overall traffic system resilience. This objective emphasizes improving velocity tracking performance through dynamic regulation of inter-vehicle spacing. Meanwhile, driving comfort, represented by $J_i^c = \| \triangle v_i(t+\tau|t)\|_F^2$, focuses on optimizing the user's driving experience by promoting smooth and progressive speed transitions. The goal of this item is to reduce discomfort and jolts during acceleration and braking processes.

Furthermore, MBRL leverages a predictive model of the environment to forecast its actions, enabling a more systematic approach to reward function design. The predictive capability allows the agent to evaluate potential future states and their associated rewards, facilitating the identification of reward structures that promote desirable long-term behaviors. Therefore, the reward function could be rewritten as

$$r_i(t) = -\sum_{\tau=0}^{L-1} \left(\| \triangle h_i(t+\tau|t)\|_Q^2 + \| \triangle v_i(t+\tau|t)\|_F^2 + \|a_i(t+\tau|t)\|_R^2 \right) \tag{14}$$

where Q, F and R are the position constant value. In addition, the action input a should also stay within a reasonable range ($a_{min} \le a \le a_{max}$) to further improve driving comfort.

Although MBRL allows for more complex reward structures by predicting long-term outcomes, these functions must be carefully balanced to avoid overfitting the modelling of simulated experiences.

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Algorithm 1 Policy Training Process of TMBRL

Input: Policy network π_θ, Environment simulator E in the source domains

Output: Trained policy network $\pi_{\theta_{\text{train}}}$ in the source domains

Initialize network π_θ, surrogate model p_ψ, empty environment replay buffer $\mathcal{B}_{env}$, model replay buffer $\mathcal{B}_{model}$ and environment simulator E
for $t = 1, \cdots, L$ **do**
  Execute action $a_t \sim \pi_\theta(\cdot|s_t)$ to obtain reward r_t, and observe next state s_{t+1}
  Store transition policy (s_t, a_t, r_t, s_{t+1}) to buffer $\mathcal{B}_{env}$
  Train model p_ψ using $\mathcal{B}_{env}$ via multi-step loss
end for
for $t = 1, \cdots, N_c$ **do**
  Sample a batch uniformly from buffer $\mathcal{B}_{env}$
  Perform k-step rollouts with policy π_θ
  Store transition to the buffer $\mathcal{B}_{model}$
end for
for updates $t = 1, \cdots, N_b$ **do**
  Select a mini-batch $\mathcal{B} = \{(s, a, r, s')\}$ from $\mathcal{B}_{env}$ and $\mathcal{B}_{model}$
  Optimize policy parameters π_θ with gradient ascent
  Adjust sample ratio $P[b \subseteq B_{model}]$
end for
return Policy network $\pi_{\theta_{\text{train}}}$ of the source domain

C. Policy Training

The primary goal of policy training is to optimize the policy, enabling the agent to choose actions in each state that maximize cumulative rewards. However, the policy training process must carefully balance between model-based predictions and real-world feedback to avoid the overfiting of inherent inaccuracies.

The framework and process of the proposed MBRL algorithm for the training policy are outlined in Algorithm 1.

1) **Model Initialization:** Randomly initialize the critic network $Q_{\lambda_i}, i = 1, \ldots, L$, surrogate model p_ψ and actor network π_θ. Initialize empty environment transitions buffer $\mathcal{B}_{env}$, surrogate model transitions buffer $\mathcal{B}_{model}$.
2) **Data Collection:** The environmental state s_t is perceived by the agent. The agent formulates a Gaussian policy and takes an action a_t, resulting in a reward r_t as the environment transitions to the next state s_{t+1}. Then, the transition policy (s_t, a_t, r_t, s_{t+1}) is stored to $\mathcal{B}_e nv$.
3) **Model Rollouts:** This process requires train dynamic model p_ψ using $\mathcal{B}_e nv$ with multistep loss. Once the dynamics model has reached convergence, a batch of transition samples can be randomly selected from the environment buffer $\mathcal{B}_{env}$. The models carry out rollouts for k steps in the batch, generating model transitions that are subsequently stored in the model buffer $\mathcal{B}_{model}$.
4) **Policy Training:** A mini-batch of transitions $\mathcal{B} = \{(s, a, r, s')\}$ is randomly sampled from $\mathcal{B}_{env}$ and $\mathcal{B}_{model}$ based on the sampling ratio $\Pr[b \subset \mathcal{B}_{model}]$. Subsequently, the network Q_{λ_i} and policy parameters θ are updated. Finally, the sampling ratio $\Pr[b \subset \mathcal{B}_{model}]$ is adjusted to enhance performance.

Algorithm 2 Policy Transfer Process of TMBRL

Input: Policy network $\pi_{\theta_{\text{train}}}$ of the source domain, replay buffer $\mathcal{B}$

Output: Optimal policy π^*_θ of the target domain

Initialize the policy networks $\pi_{\theta_{\text{train}}}$ with weights θ^*_M, replay buffer $\mathcal{B}$
Obtain initial state s_t from the target domain
for episode = 1, ..., L **do**
  Map source state-action pairs to target using the transformation function
  Determine action $a_t = \pi^*_\theta(s_t \mid \theta) + \mathcal{N}_t$ in terms of the current policy and exploration noise
  Execute action a_t on the target environment, and collect the reward r_t along with next state s_{t+1}
  Store transition (s_t, a_t, r_t, s_{t+1}) in the buffer $\mathcal{B}$
  Sample a minibatch of transitions from buffer $\mathcal{B}$
  Update the policy network π_θ
end for
return Optimal policy π^*_θ of the target domain

D. Policy Transfer

The Algorithm 2 outlines a TMBRL process for vehicle platoon. It starts by initializing the policy networks with specific weights and setting up a replay buffer. The initial state from the target domain is acquired and used across multiple episodes. During each episode, actions are determined by the current policy and exploration noise, prompting interactions with the environment that yield rewards and new states. Transitions are stored in a replay buffer for random minibatch sampling, optimizing the policy network via a reinforcement learning algorithm. Finally, the network is updated iteratively to achieve the optimal policy for efficient vehicle platooning in the target domain.

With this in mind, the principles of transfer learning applied in vehicle platoon are detailed in Fig. 2. To be more special, the knowledge from CAV_1, comprising transition policy (s_t, a_t, r_t, s_{t+1}), serves as the source domain. Then, the knowledge is transferred to the other vehicles (CAV_i, $i \geq 2$), denote as the target domain. It should be noted that target domain vehicles only need to update the knowledge according to the individual dynamic characteristics and tasks. In this way, the initial optimization process of algorithm could be omitted, thereby enhancing the convergence performance of platoon.

IV. Experiments Design and Results

A. Experimental Platform

The experimental validation is conducted on a vehicular platoon co-simulation platform (refer to Fig. 2), utilizing Prescan® software in conjunction with a high-performance computing system. The computing hardware comprises an Intel i7-14700KF CPU operating at 3.4GHz and an NVIDIA RTX4060 GPU. Prescan® serves as a comprehensive simulation environment for autonomous vehicle testing, featuring

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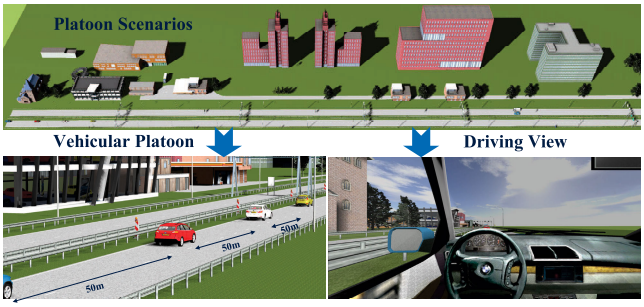

Fig. 3. Platoon scenarios.

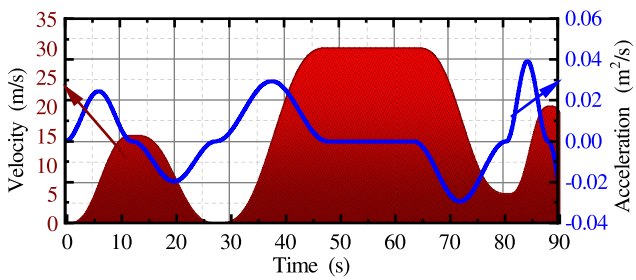

Fig. 4. Anticipant velocity and acceleration.

advanced capabilities in realistic scenario construction and multi-sensor system integration.

The proposed TMBRL strategy is deployed on the host computer using Python, facilitating the real-time generation of optimal control actions (Acceleration). In addition, the spacing and velocity information of the ahead vehicle could be obtained through the camera tool from Prescan software.

B. Experimental Scenarios

As illustrated in Fig. 3, the platoon configuration with three vehicles traveling on a 2km urban traffic scenario. Three default vehicle models are chosen from Prescan®, and the leading vehicle. According to the property of safety driving, the desired safe spacing of platoon is set as 50m. The anticipant velocity and acceleration are set, as shown in Fig. 4. This graph simulates various traffic scenarios in a vehicular platoon environment.

C. Experimental Setup

The proposed TMBRL strategy is evaluated by comparing different benchmarks. It is important to note that each training episode is terminated if any of the predefined constraints, such as operational limits on system performance or safety thresholds, are violated. In the training process, episodes are terminated if any of these operational limits are violated, to prevent unsafe or unrealistic behaviors from emerging during the learning process. Moreover, several experimental parameters are given in TABLE I.

Additionally, three baselines are conducted to validate the performance of the proposed strategy, detailed as follows

- **Experiment 1**. The model-free reinforcement learning with deep deterministic policy gradient (MFRL-DDPG) strategy is implemented in the vehicular platoon control [30]. DDPG employs an actor–critic framework in which the actor network outputs a deterministic continuous action for each state, while the critic evaluates the corresponding Q-values.
- **Experiment 2**. The model-free reinforcement learning with deep Q-network (MFRL-DQN) strategy is implemented for platoon control [31]. DQN is a value-based, off-policy method that estimates action values to inform decision-making in discrete action spaces, such as choosing between different levels of acceleration or deceleration. Here, DQN could be used to balance exploration and exploitation as vehicles interact with the environment.
- **Experiment 3**. The model-based reinforcement learning with model-based policy optimization (MBRL-MBPO) strategy is implemented to platoon control [10]. MBPO enhances sample efficiency by learning a predictive model of the environment, which simulates future vehicle trajectories based on current policies. This allows for the generation of synthetic data, which can be used to train the reinforcement learning policy without requiring as much environmental interaction.
- **Experiment 4**. The proposed TMBRL-MBPO strategy is implemented to platoon control. Different from **Experiment 3**, the transfer learning method entails deploying the surrogate model from a source domain to a target domain. The surrogate model is fine-tuned on the new domain, reducing the need for extensive retraining. This transfer learning approach allows for faster adaptation to new traffic scenarios and improves computational efficiency, making it particularly useful for deploying platoon control in real-world settings with limited training data.

TABLE I
EXPERIMENTAL PARAMETERS

Symbol	Description	Value
T_s	Sampling frequency	$100Hz$
γ	Discount factor	0.98
N_c	Number of neurons	65
L	Predictive horizon length	20
N_e	Experience buffer length	1e6
N_b	Mini batch size	64
E_{max}	Max episodes	200
α	Temperature parameter	1.2
β	Learning rate	0.001
$v_{\max}$	Maximum velocity	35m/s

V. RESULTS AND DISCUSSIONS

This section presents a comprehensive analysis of the training performance of the proposed algorithms and evaluates their effectiveness in the context of platoon tracking applications.

A. Verification of Ploicy Training

The average reward value serves as a crucial metric for evaluating the performance of reinforcement learning algorithms. Higher average rewards indicate better decision-making capabilities and more efficient learning. Fig. 5 reveals the

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TABLE II
EXPERIMENTAL RESULTS ANALYSIS WITH AMAXE AND RMSE

Symbol	MAE			RMSE			R^2		
	$\triangle h$(m)	$\triangle v$(m/s)	a(m/s^2)	$\triangle h$(m)	$\triangle v$(m/s)	a(m/s^2)	$\triangle h$(m)	$\triangle v$(m/s)	a(m/s^2)
MFRL-DDPG	21.295	7.952	0.081	12.208	3.623	0.0097	0.899	0.865	0.615
MFRL-DQN	10.193	3.593	0.052	5.207	1.276	0.0094	0.989	0.985	0.605
MBRL-MBPO	8.275	1.771	0.025	4.416	0.778	0.0038	0.991	0.997	0.937
TMBRL-MBPO	7.382	1.149	0.055	3.987	0.519	0.0036	0.996	0.994	0.992

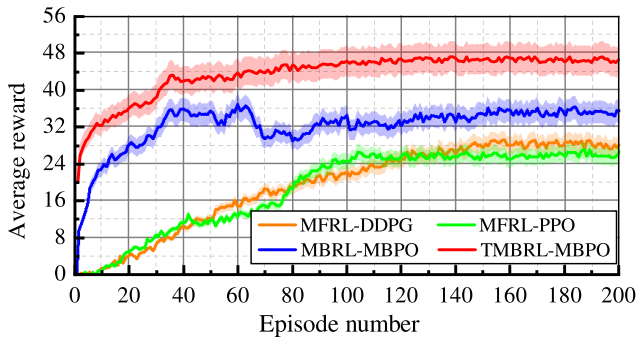

Fig. 5. Average reward with different strategies.

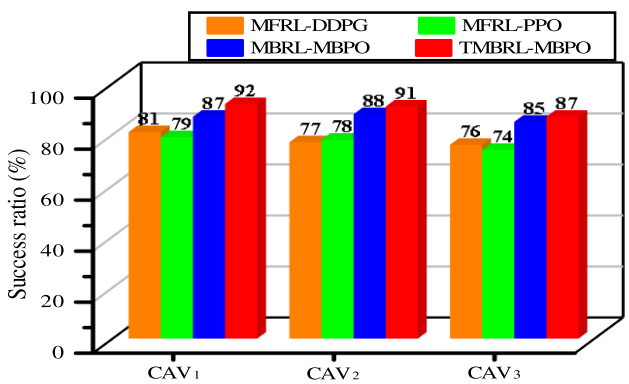

Fig. 6. Success ratio with different strategies.

distinct superiority of the TMPRL-MBPO strategy compared to other reinforcement learning methods. Specifically, the MFRL-DDPG and MFRL-DQN demonstrate lower average rewards, albeit with a noticeable upward trend, and eventually reaching 27.5 and 26.1. This phenomenon is attributed to their reliance on trial-and-error exploration without leveraging surrogate models, which limits their ability to anticipate future states and adapt policies efficiently. The MBRL-MBPO exhibits obvious advancement but the peak performance is limited to 37. In contrast, the TMBRL-MBPO inherits the network policy from MBRL-MBPO, resulting in a higher initial reward of around 20. Accordingly, the reward value consistently improves and converges to approximately 47, showcasing enhanced stability and efficiency across episodes. This is because that TMPRL-MBPO integrates transfer learning within the platoon, enabling the efficient reuse and adaptation of learned knowledge across different tasks.

In reinforcement learning applications, the success rate could assess the proportion of tasks completed successfully, indicating the model's effectiveness and generalization capabilities. Fig. 6 illustrates a comparative analysis of success ratios across three vehicles (CAV_1, CAV_2 and CAV_3) utilizing four reinforcement learning methodologies. Notably, TMBRL-MBPO outperforms the other approaches, achieving the highest success rates of 92%, 91% and 87%. MBRL-MBPO follows closely with slightly lower success ratios with 79%, 78% and 76%, benefiting from its competitive success rates by ensuring efficient vehicle spacing and speed adjustments in complex environments. In contrast, both MFRL-DDPG and MFRL-DQN exhibit comparatively lower success rates. It reflects the limitations of model-free strategies in handling complex environmental dynamics without forward-looking prediction and adaptation mechanisms.

B. Verification of Platoon Tracking

The platoon tracking performance is evaluated in terms of the spacing and velocity. Fig. 7 presents the results of velocity responses of CAV_1, CAV_2 and CAV_3, which are subject to different control strategies. According to Fig. 7(a), MFRL-DDPG exhibits a velocity error spike peaking at approximately 16 m/s initially, with fluctuations persisting throughout the duration. Fig. 7(b) reveals MFRL-DQN towards more pronounced fluctuations, with errors peaking near 6.5 m/s, reflecting challenges in real-time adaptation. Fig. 7(c) for MBRL-MBPO shows a peak velocity error of about 3m/s, yet with more rapid stabilization over time due to surrogate modeling support. Most notably, Fig. 7(d) illustrates the performance of TMPRL-MBPO method, maintaining a relatively stable velocity error with peaks reaching around 1.8m/s, followed by a rapid convergence. This reflects the method's strength in applying transfer learning and predictive capabilities, allowing for efficient adaptation and error management. Overall, these quantitative results emphasize the superiority of TMPRL-MBPO in handling dynamic changes with minimal error.

The results of spacing error with different strategies are analyzed in Fig. 8. As depicted in Fig. 8(a), MFRL-DDPG reaches a peak spacing error of about 28m around 60s, with gradual improvement thereafter. Fig. 8(b) presents the performance of MFRL-DQN, where errors increase to approximately 19m, indicating the limitation in immediate adaptation. MBRL-MBPO, depicted in Fig. 8(c), initially experiences errors up

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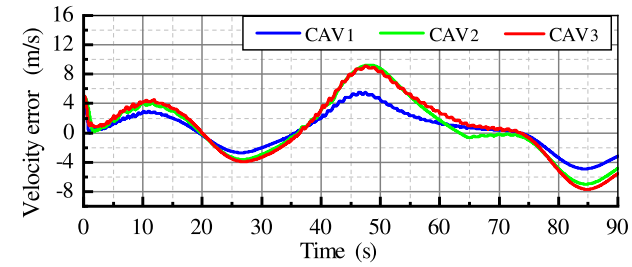
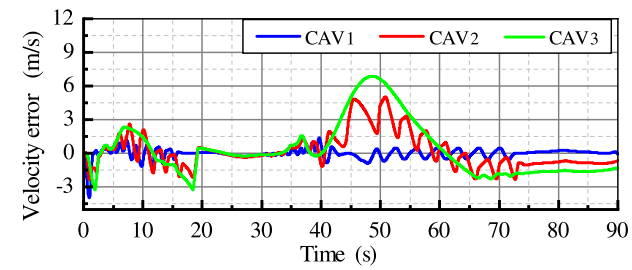
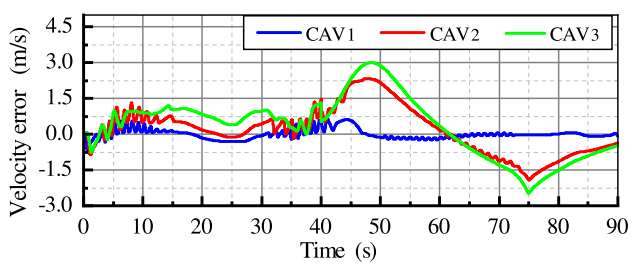


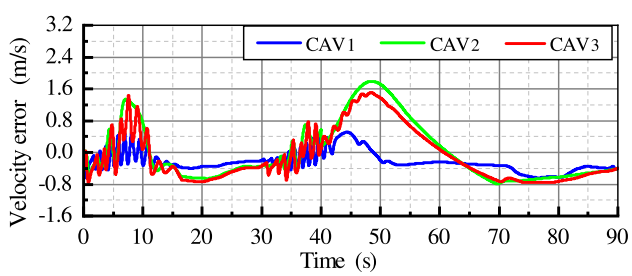
(a) Velocity error of MFRL-DDPG



(b) Velocity error of MFRL-DQN

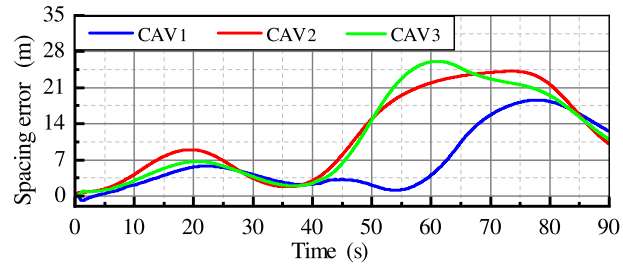


(c) Velocity error of MBRL-MBPO

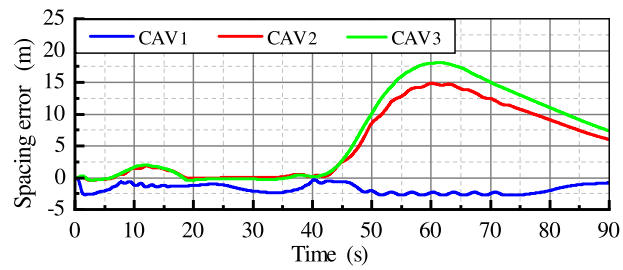


(d) Velocity error of MBRL-MBPO

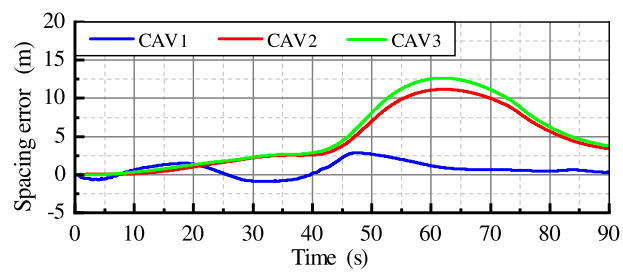
Fig. 7. Results of the velocity error with different strategies.



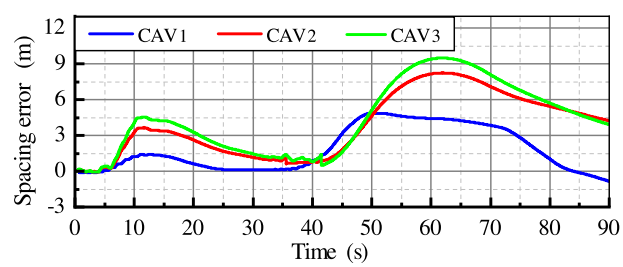
(a) Spacing error of MFRL-DDPG



(b) Spacing error of MFRL-DQN



(c) Spacing error of MBRL-MBPO



(d) Spacing error of MBRL-MBPO

Fig. 8. Results of the spacing error with different strategies.

to 13m but benefits from predictive adjustments, leading to more effective error stabilization. In contrast, TMPRL-MBPO starts with an error of 10m and rapidly reduces these errors in Fig. 8(d). It indicates TMPRL-MBPO approach effectively minimizes error impact, showcasing its superiority in adapting to dynamic vehicular conditions.

The acceleration results with different control strategies are presented in Fig. 9. Both MFRL-DDPG and MFRL-DQN suffer from significant variability and wider boundary values, suggesting frequent fluctuations and occasional instability. Conversely, MBRL-MBPO achieves better control with reduced variability, reflecting enhanced stability of platoon. Notably, TMPRL-MBPO outperforms the others by demonstrating tightly controlled variability boundary extremes, underscoring its superior robustness in managing the acceleration of platoon.

To further elaborate on the advantages of our proposed TMBRL strategy, the maximum absolute error (MAE) and root mean square error (RMSE) assessment indexes are introduced. With this in mind, the spacing error ($\triangle h$), velocity error ($\triangle v$) and acceleration (a) of platoon are presented in TABLE II. MFRL-DDPG shows the highest errors, especially in acceleration control. MFRL-DQN reduces these errors but still struggles with accuracy. MBRL-MBPO improves performance with lower errors and a notable coefficient of determination (R^2) of 0.937 in acceleration, indicating a strong model fit.

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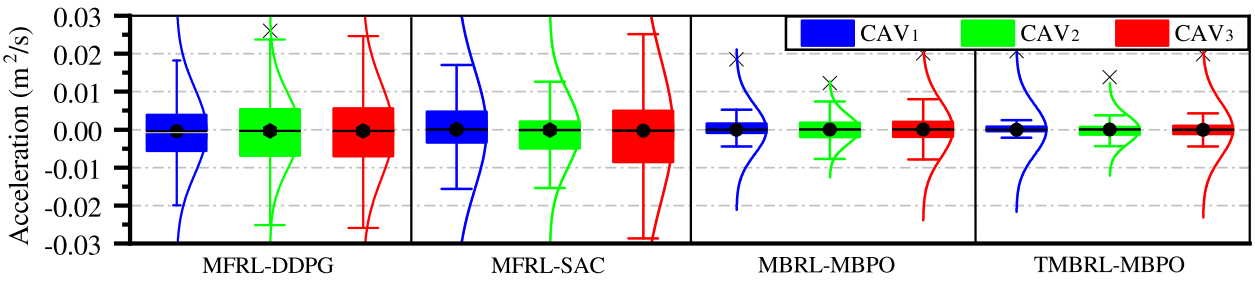

Fig. 9. Results of the acceleration with different strategies.

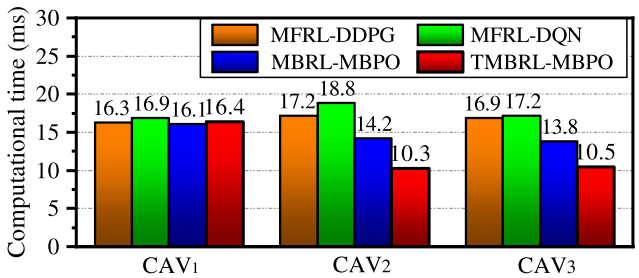

Fig. 10. Results of computational time with different strategies.

TMPRL-MBPO achieves the lowest errors and near-perfect R^2 values, highlighting its superior precision and robustness in managing vehicular platoon.

C. Verification of Computational Time

Fig. 10 shows the average computational time of four strategies across the platoon. Notably, TMBRL-MBPO outperforms the other methods in terms of computational efficiency. Here, CAV_1 represents the source domain vehicle, its computational efficiency remains largely consistent across all strategies. For CAV_2, TMBRL-MBPO requires only 10.3 ms, compared to 14.2 ms for MBRL-MBPO, 17.2 ms for MFRL-DDPG, and 18.8 ms for MFRL-DQN, representing a 27.4% improvement over MBRL-MBPO. In CAV_3, TMBRL-MBPO achieves 10.5 ms, outperforming MFRL-DDPG (16.9 ms) by 4.5%. These results show that MBRL methods, such as TMBRL-MBPO, are more computationally efficient than model-free approaches like MFRL-DDPG and MFRL-DQN. Additionally, transfer learning significantly improves the efficiency of TMBRL-MBPO, particularly in target-domain vehicles, highlighting its potential for reducing computational demands.

Therefore, it can be concluded that the TMPRL-MBPO strategy could maintain the platoon tracking performance while improving the efficiency of policy learning.

VI. Conclusion

In this paper, we have developed a TMBRL strategy to improve the sample efficiency and learning speed for CAVs. An MBRL algorithm involving surrogate model and SAC method has been constructed. The surrogate model serves as an approximate representation of the environmental dynamics, capturing the state transitions that occur within the platooning scenario. SAC was characterized by its entropy maximization objective, which promotes diverse exploration of the action space. To enhance the computational efficiency of the platoon control algorithm, we have employed a transfer learning approach within the MBRL framework. Finally, the experimental results demonstrated that our proposed TMBRL strategy could provide satisfactory platoon tracking performance and average reward. In this paper, limited experimental scenarios and datasets make it challenging to validate the generalizability of the proposed strategy on complex scenarios. In future work, we will focus on testing the proposed strategy in actual vehicles.

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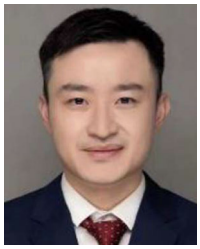

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